Class Meeting Handout #14: Comments

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

In this document, we will present the plots requested in the activities of the handout.[1]

[1] Read chapter 8.3 and 8.4.

Exercise #1: Pre-class Activity

1. Compare the plots that you prepared using the data from the pre-class exercise with your peers. What did you observe?

Using the data tables provided in the handout and in the pre-class activity, let us explore the data plots and interpret their meaning. You were asked to plot data for the base-catalyzed hydrolysis of a series of substituted ethyl benzoate esters. In this case, we broke Hammett's rule and included *ortho* substituents in the data set.

Using the data set in Table 1 of the instructions for the pre-class activity, a plot of the data set was generated and is presented in Figure 1. As you can see, it appears that there is a poor correlation.

Hammett excluded *ortho*-substituted benzoates; let us do the same. A plot that uses only the *meta* and *para*-substituted reactants is presented in Figure 2.

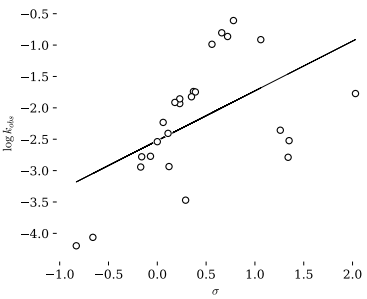


Figure 1: Plot of base-catalyzed hydrolysis of ethyl benzoates. $\rho = 1.0 \pm 0.3$, $r^2 = 0.33$. ↓

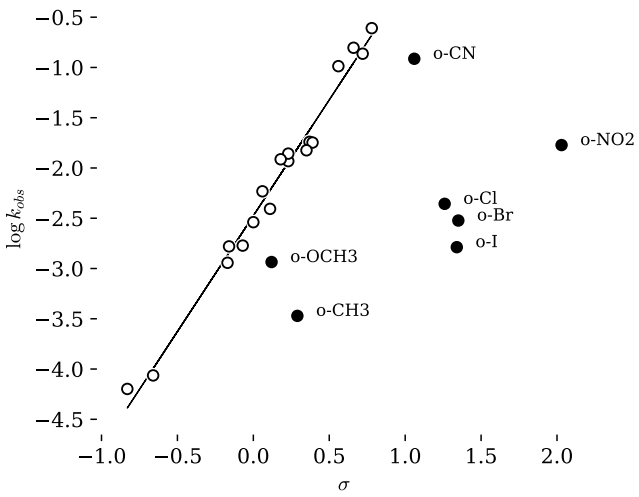


Figure 2: Plot of base-catalyzed hydrolysis of *meta* and *para*-substituted ethyl benzoates (open circles). ←
$\rho = 2.30 \pm 0.07$, $r^2 = 0.983$.
Results with *ortho*-substituted benzoates that were excluded are shown as dark circles.

The plots in figures 1 and 2 were produced using the *Python* notebook accessible at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_17/4410_15-PreClassActivityPlot.ipynb

Exercises #2 and #3: Hammett Plots of Acid and Base Hydrolysis

2. Plot the data for acid hydrolysis of substituted ethyl benzoates presented in table 1 against the Hammett σ values. What is the reaction constant, ρ, for acid-catalyzed benzoate ester hydrolysis? Is there a significant substituent electronic effect in this process?

3. Taft used the reaction constant for base-catalyzed hydrolysis of substituted ethyl benzoates as the "standard ρ^* value" for all ester hydrolysis reactions. Plot rate data for this reaction and determine the reaction constant, ρ^*. We will use this value as our standard value. Table 2 in the class meeting handout collects some rate data from Ingold and Evans.

The two data sets from tables 1 and 2 in the class meeting handout are plotted in Figure 3 below.[2] We see that the acid-catalyzed reaction has little sensitivity to the electronic effect of substituents while the base-catalyzed reaction is very sensitive.

[2] The plots in Figures 3 and 4 were produced using the *Python* notebook accessible at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_17/17-ClassExample-table1and3.ipynb

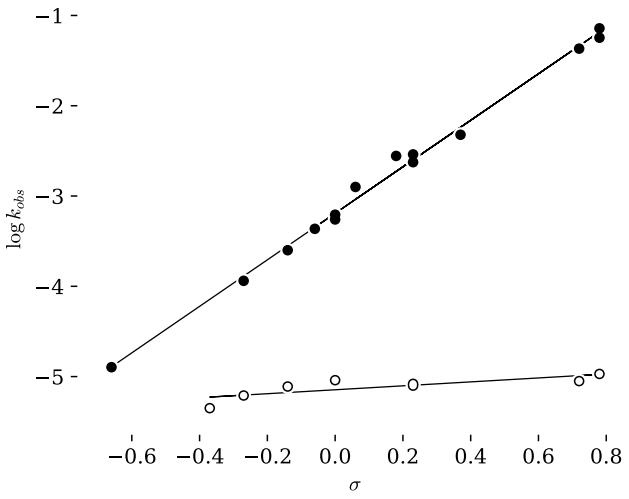


Figure 3: Plot of acid and base-catalyzed hydrolysis of *meta* and *para*-substituted ethyl benzoates.
←
The acid-catalyzed reaction (open circles) is not sensitive to electronic substituent effects. $\rho = 0.22 \pm 0.07$, $r^2 = 0.637$ (nearly horizontal correlations have low values for r^2.)

The base-catalyzed reaction (closed circles) is sensitive to electronic substituent effects. $\rho = 2.58 \pm 0.05$, $r^2 = 0.994$

Did you plot the acid-catalyzed data with the result for *ortho*-nitrobenoate included? That was a trap. You can see that the steric effect of the *ortho* takes the point far off the line established by traditional Hammett substituents as shown in Figure 4.

Figure 4: The same plot as in figure 3 with the value for *ortho*-nitrobenoate included.↓

